

MDS5202 Assignment 5: Model selection, regularization, and dimension reduction

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Conceptual

1. We will now derive the Bayesian connection to the lasso and ridge regression discussed in Section 6.2.2.

- (a) Suppose that

$$y_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \epsilon_i,$$

where $\epsilon_1, \dots, \epsilon_n$ are independent and identically distributed from a $N(0, \sigma^2)$ distribution. Write out the likelihood for the data.

- (b) Assume the following prior for β : $\beta_1, \dots, \beta_p$ are independent and identically distributed according to a double-exponential distribution with mean 0 and common scale parameter b ; i.e.

$$p(\beta) = \frac{1}{2b} \exp(-|\beta|/b).$$

Write out the posterior for β in this setting.

- (c) Argue that the lasso estimate is the *mode* for β under this posterior distribution.
- (d) Now assume the following prior for β : $\beta_1, \dots, \beta_p$ are independent and identically distributed according to a normal distribution with mean zero and variance c . Write out the posterior for β in this setting.
- (e) Argue that the ridge regression estimate is both the *mode* and the *mean* for β under this posterior distribution.

Applied

2. In this exercise, we will predict the number of applications received using the other variables in the **College** data set.
 - (a) Split the data set into a training set and a test set.
 - (b) Fit a linear model using least squares on the training set, and report the test error obtained.
 - (c) Fit a ridge regression model on the training set, with λ chosen by cross-validation. Report the test error obtained.
 - (d) Fit a lasso model on the training set, with λ chosen by cross-validation. Report the test error obtained, along with the number of non-zero coefficient estimates.

- (e) Fit a PCR model on the training set, with M chosen by cross-validation. Report the test error obtained, along with the value of M selected by cross-validation.
- (f) Fit a PLS model on the training set, with M chosen by cross-validation. Report the test error obtained, along with the value of M selected by cross-validation.
- (g) Comment on the results obtained. How accurately can we predict the number of college applications received? Is there much difference among the test errors resulting from these five approaches?